

Chain-of-Thought Faithfulness With Ground Truth

504 agent trajectories where the correct answer is known independently of the model, and a headline result that does not survive de-duplication

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Abstract

Agentic harnesses for vulnerability localisation emit long reasoning traces and then reduce them to a ranked list of files, discarding the reasoning. Whether that discarded text carries information the structured output does not is a chain-of-thought faithfulness question, and unusually it is one with an answer key: each task is a real CVE at the commit before its fix, so the fix commit names the true files independently of anything the model said. We mine 504 trajectories for *relocations* — statements where a run contradicts the question it was given and names a different file — report that our own headline result does not survive de-duplication, and then ask at the activation level whether the relocated file, or the decision to relocate at all, is recoverable before the model writes it. Relocations are extracted with four filters (position within the step, supersession, a real-extension test, and hedge detection), which remove 33 of 133 raw candidates on a held-out sample; every readout is scored a second time against a seeded decoy, every probe is cross-validated leave-one-claim-out with dimensionality reduction fitted inside each fold, and every AUC is reported against a permutation null for the identical procedure rather than against 0.5. Counted per mention, relocations name a true gold file 81.4% of the time, apparently beating the 49.8% calibration of three-of-three pass agreement; counted per distinct (task, file) claim — the unit the baseline is measured on — precision is 40.5% on 37 claims, below the baseline, with three tasks contributing over half the population and one a single correct claim restated 55 times; and at the token before the file is named, asking the model outright recovers it in 18 of 30 claims against a decoy floor of 3 ($p = 0.0003$) while a Jacobian lens recovers 11 of 31 against a floor of 5 ($p = 0.15$). The activation-level follow-ups are weaker still and are reported as such: a trained readout appears to recover the file at 20 of 24 claims, but relocated filenames are nested inside repositories and the deconfounded version scores two; right-versus-wrong relocation is null with a detection threshold of AUC 0.676 at 35 claims; and whether a relocation is coming is suggestive at late layers while surviving no correction for multiple comparisons. For trace mining specifically, the unit of analysis determines the headline, and repetition within a trace is not independent evidence. More broadly, five separate artifacts in this work each produced a publishable-looking number and every one pointed the way the hypothesis wanted — a repository detector, a delimiter detector separating classes at the embedding layer before any block had run, a prompt-length shortcut, a per-mention count, and a permutation null built at the wrong unit — none caught by finding the result implausible, and each caught by a control built in advance to catch that class of error.

Keywords chain-of-thought faithfulness, mechanistic interpretability, LLM evaluation, agentic harnesses, code localization

Confidential working draft

Circulated for individual review only. **Please do not redistribute, post, or cite.** Findings here are provisional: the pilot rests on ~ 30 distinct claims, its lens arm is

not statistically significant, and every activation-level follow-up returned a null or a corrected artifact. Five numbers were corrected during the work and each is reported with **both** the original and the corrected value rather than only the survivor. Numbers may change.

1. WHY THIS SETTING HAS GROUND TRUTH

Faithfulness experiments usually cannot check a model's stated target against an objective answer. The stated reasoning and the final output are both model artefacts; a third, independent statement of what the answer *is* rarely exists.

Vulnerability localisation supplies one. A task is a repository checked out at the parent of a known fix commit. The weakness is present at that commit and absent at its child. The fix commit is a diff, so it names the files and line ranges that were actually wrong, and it was written by the project's own maintainers rather than by an annotator reading model output.

The model is given a symptom, the repository, and nothing else — no path, no CWE identifier, no advisory text. It explores with `glob`, `grep` and `read`, then submits files and cites lines. Its transcript, its structured output, and the true answer are three independent objects.

This yields a triple per trace: what the model **said**, what it **submitted**, and what was **true**. Faithfulness questions that are normally circular become measurable.

The corpus is 594 tasks drawn from real advisories, which produced 1,288 casefiles and 504 recorded reasoning trajectories.

2. APPARATUS

Three models with different jobs, on one workstation, with two model-free retrieval stages and a mechanical evidence stage that uses no model at all.

stage	model	job
locate	Antares-1B	ranks candidate files; run k times per task for agreement
plan / judge	Qwen3.6-35B-A3B	selects files to examine, scores
enrich	FastContext-4B	explores inside a chosen file and cites lines
evidence	—	resolves every claim against the real checkout

Everything is local, so full activations are reachable for the follow-up proposed in §7. Every run writes an ATIF trajectory (per-node step records, tool calls, reasoning text) and a casefile (the structured output, ranked candidates, cited line ranges).

Locate is run three times per task. The number of passes that name a file calibrates how likely it is to be gold: 1 of 3 arrow.r 8.0%, 2 of 3 arrow.r 22.6%, 3 of 3 arrow.r

49.8%, measured on this corpus. That calibration is the baseline any mined signal has to beat, and §4 turns on the fact that it is measured per distinct file, not per mention.

3. RELOCATIONS

A **relocation** is a statement in the reasoning where a run contradicts the question it was given and names a different file than the one it was sent to examine. It is the simplest case where the transcript and the structured output can disagree about a checkable fact.

Most apparent contradictions are not contradictions. Four filters, validated on a held-out sample where they removed 33 of 133 raw candidates:

1. **Position within the step.** A claim made mid-reasoning and abandoned before the step ends is not the run's position.
2. **Supersession.** A claim contradicted by a later claim in the same trace is superseded, not held.
3. **Real-extension test.** Naming an additional file is an extension, not a contradiction, unless it displaces the original.
4. **Hedge detection.** "This may live in X instead" is not an assertion that it does.

192 relocations survive across the corpus, covering 37 distinct (task, file) claims.

Those two numbers are the same evidence counted two ways.

4. THE RESULT, AND ITS RETRACTION

We measured whether a relocation names a true gold file, and compared it against the three-pass agreement calibration.

unit	n	precision	vs. 49.8% baseline
per mention	192	81.4%	above
per distinct (task, file) claim	37	40.5%	below

Same data, same filters, opposite conclusions. The per-mention figure was our headline. It is wrong, for a reason that is visible once the population is decomposed:

- Three tasks contribute more than half of all mentions.
- One of them is a single correct claim restated 55 times within one trace.

A model that returns to the same conclusion as it works produces many mentions of one claim. Per mention, that reads as fifty-five successes. The baseline it was being compared against is measured per distinct file, so the comparison was never like-for-like.

The correct unit is the claim, because the claim is what a downstream consumer would act on: a file is either submitted or not, however many times the reasoning

mentioned it. On that unit the mined signal is **below** the agreement baseline, and does not justify being used as a filter.

This generalises past this harness. Any pipeline that mines assertions from long transcripts can count mentions, and mentions are the default unit because they are what a regex returns. The inflated number is the one that looks publishable.

5. WHAT SURVIVED

The extraction is sound, and one finding is unaffected by the counting error because it is a question about presence rather than precision:

Of 218 relocations mined from reasoning, **6** fail to appear in the structured output.

97.2% of what the model states in its reasoning does reach the ranked list. The reduction from transcript to structured output is therefore not discarding the model's stated conclusions. This is a negative result for the "reasoning is thrown away" hypothesis in this setting.

But the six are not random, and finding out why changes what they mean.

Split by the date the enricher prompt changed:

	relocations	never reached the output	
before 2026-08-31	29	5	17.2%
on or after	189	1	0.5%

Fisher exact, two-sided: **p = 0.00016**, odds ratio 39. The earlier prompt carried a consolation clause — "*then cite the closest line in the file you were asked about*" — and one trace shows the model walking straight into it:

"But the actual vulnerability (the default verify=False) is in http_hook.py:163. I should cite that file and note it. Wait, the instructions say: 'If what you find says the weakness is really in another file, SAY SO in your explanation and name it — that is worth recording — then cite the closest...'"

It located the weakness, resolved to cite it, and was redirected by the instruction into filing an in-scope citation instead. That clause was removed on 2026-08-31, and the effect went with it.

So the answer to the third follow-up question below is neither of the two it proposes. The never-submitted set is not a suppression story and not an error story — it is instruction following, and the instruction was ours. The honest reading of the 97% is therefore that it measures a prompt as much as a reduction: under a prompt that asked for a consolation citation, one relocation in six was lost.

Whether those conclusions are *weighted* correctly is a separate question. On current evidence we do not know, and §4 is the reason we decline to claim otherwise.

6. INSTRUMENT VALIDATION

Claims about a model are only worth their instrument. Two mechanisms carry that weight here.

Every claim is resolved against the real checkout on the way into a casefile. A path that does not exist at that commit is struck through, scored `p_gold 0.0`, and labelled fabricated rather than dropped, so the failure stays visible. This fires on 3 of 20 tasks in the current bank — fabricated paths are built out of the question’s own vocabulary, which makes them the most plausible-looking rows in the table.

Measurements were audited against their own inputs. Eight were wrong before they were checked. Each was found by asking a number what it was made of:

reported	actual
Parser read 1,288 / 1,288 casefiles	Candidates found in 437; 831 used a five-column table the positional regex did not match
Enrichment is a no-op across four interventions	A dropped range in aggregation; the intervention was never measured
Model served with reasoning disabled	Three flags asserted it; 107 of 107 streamed tokens were reasoning
Arms differ by enricher	Locate was not replayed, so the arms received different file lists

The last two are recent. The third is instructive: a server flag named for the desired behaviour, accepted without error, that did not produce it — the request-level parameter overrode it. A flag named for the thing you want is not evidence the thing happened.

Arms are now run against a replayed locate stage, so the file list is identical across arms and the enricher is the only variable.

7. PILOT: IS THE RELOCATED FILE DECODABLE BEFORE IT IS STATED?

The follow-up below is no longer hypothetical. Two methods were put to the same question on `qwen3.6-35b`, the model in this corpus that both relocates often (1.10 per trace, four times any other) and has a published Jacobian lens.

Each case replays a real trajectory to the moment **before** the model names the relocated file. Two cut points test different strengths of claim: `at_path` stops mid-sentence, after `vulnerability is present in ``; `at_sentence` stops before the clause exists at all. Every readout is scored a second time against a **decoy** — another case’s target, so the filename distribution matches and the answer is wrong.

Only the cut that is genuinely earlier counts. `prefix_at_sentence` falls back to the whole step when a claim has no earlier sentence boundary, and for a minority of claims it is byte-identical to `prefix_at_path` — where the model has already written in ```. Measuring “before the sentence” on those is measuring after it. Every figure below is reported on the genuinely-earlier subset, and the difference this makes is large: on the first 30-case export it moved the lens from 9/17 ($p = 0.109$) to 4/12 ($p = 0.6875$).

7.a. *The cheap baseline beats the lens*

genuinely earlier cut	claims	target	decoy	p (McNemar exact)
ask the model outright	30	18	3	0.0003
Jacobian lens, any-token scoring	31	11	5	0.146
Jacobian lens, distinctive tokens	31	11	5	0.146

The lens row is given under both scoring rules because the obvious objection is that it loses only because the permissive rule inflates its decoy. It does not: at this cut the two rules give the same 11 of 31 against the same floor of 5. The shared-token contamination is real but lives entirely at the later cut, where correcting it drops the decoy from 21 of 31 to 4.

Asking beats the lens

Same question, same corpus, same decoys: can the file be named before it is written?



Figure 1: The two methods with their decoy floors drawn across the bars rather than beside them, because a hit rate here is unreadable without the floor it has to clear.

Scored only on claims whose cut is genuinely earlier than the naming sentence.

Asking is simply appending “*which file are you about to name? Reply with one path*” to the same prefix and reading the reply. It recovers the file in 18 of 30 claims against a decoy floor of 3. The lens, on the same question and the same corpus, does not separate from its decoy.

What that decoy floor is actually worth. Building the follow-up probes surfaced a weakness in the control itself. The decoy is drawn from the whole corpus, but relocated filenames are almost perfectly nested inside repositories: 20 of the 21 distinct basenames occur in exactly one repo, and **71 of 75 decoys name a file from a different project than the one being read**. Every method here knows which project it is reading — for the baseline the transcript is in the prompt, for the lens and the probes the residual encodes it — so preferring the target over that decoy can be satisfied without ever distinguishing files *within* a repository.

The decoy column is therefore a floor that is too low rather than a noise estimate. The number that survives is the target column: 18 of 30 claims naming the exact repository-relative path, against the many files in that repository the model could have named instead. The lens is unaffected in direction — a floor that is too low could only have flattered it, and it was null regardless. The repair for a successor corpus is specific: draw relocation pairs from **within** a repository, so that project identity favours both candidates equally.

The information is there. The lens does not read it. That is the pilot’s result, and it is a statement about the instrument rather than about the model. It also makes the faithfulness question well-posed: something at that point in the trajectory determines the file the model is about to name, and it is recoverable — so “can a lens recover it?” has a real answer, and for this lens the answer is no.

The comparison is deliberately unfair to the lens, in the direction that makes the negative safe: the lens reads a single activation, while the baseline lets the model generate — up to 2,400 tokens, and on these cases it uses them. So the baseline shows the answer is reachable *with additional forward computation*, not that it sits in the residual stream. A baseline that strong losing would have been decisive for the lens; a baseline that strong winning only bounds what the lens failed to find.

Where the file becomes decodable

Fraction of 17 claims with the target in the top 20, by layer, distinctive tokens only



The transport reads the relocated file out of the middle of the stack; the plain logit lens reaches it only at layer 38, where it is decoding the answer the model is about to say.

That is the case for the transport doing real work. It is NOT a case for the strong claim: at_sentence peaks at 29% of claims and stays at $p = 0.109$, unchanged by distinctive-token scoring. Both panels are the same 17 claims, scored identically; only the cut point differs.

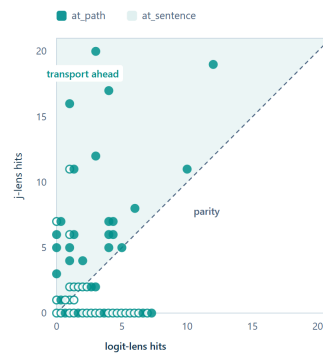
Figure 2: Where the relocated file becomes decodable, by layer, on the first 30-case export. At `at_path` the transport recovers the target from layer 20 and reaches 53% of claims by layer 30; the plain logit lens sits at or below 6% through layer 34 and arrives only at layer 38 — the final layer, where it is decoding the answer the model is about to emit. The decoy stays flat below 6%. This figure describes the later cut, where the model is already mid-sentence; it is not evidence about the earlier cut, where the lens does not beat its decoy.

7.b. What survives about the transport

Against the plain logit lens the Jacobian transport is **never worse on any of the 60 readouts** of the first export, carrying $4.2\times$ the layer-hits before the naming sentence and $2.7\times$ after. That is a real property of the transport and it is orthogonal to the result above: being a better readout than the logit lens does not make it a sufficient one.

The transport is never worse

j-lens against plain logit lens, per readout, same forward pass



0 of 60

rows where the transport does worse

4.2x

layer-hits before the sentence exists (38 against 9)

2.7x

layer-hits once it has begun (195 against 72)

16 vs 14

per CLAIM at `at_path` — the binary hides the margin

Per claim the two lenses look close at `at_path`, and the report says so. Per readout they are not close: the transport leads on 24 of 30 rows there, 10 of 30 before the sentence, and trails on none. The claim-level test is the conservative one and stays the headline.

Figure 3: Jacobian transport against the plain logit lens, per readout, same forward pass. Every point sits on or above the diagonal. Per *claim* the two look close at `at_path` — 16 of 17 against 14 of 17 — which is the conservative test and stays the headline.

7.c. The scoring defect the control found

A hit was counted when **any** token of a filename entered the top 20. Two unrelated files sharing `.py` therefore scored for the target and the decoy at once.

The decoy is silent unless it shares a token

60 readouts, split by whether decoy and target have any token in common



Voiding decoy hits that ride on a shared token moves at_sentence from $p = 0.109$ to $p = 0.0078$.
This is ONE-SIDED and is not the result: the same collision can inflate a target hit, and the log cannot say which token fired.

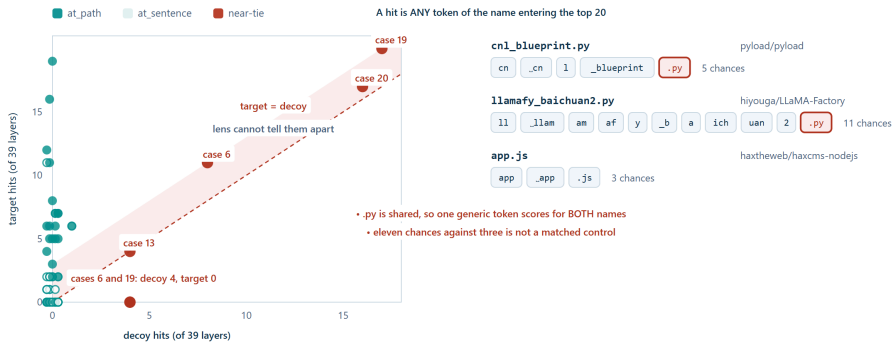
Name LENGTH is not the confound — Pearson r between token count and hits is 0.183 over 120 observations, and the 4-token name outscores the 11-token one.
Shared tokens are. The fix is to score on distinctive tokens only, symmetrically, which needs the re-run rather than a re-analysis of this log.

Figure 4: The decoy's noise floor, split by whether it shares a token with the target. Sharing nothing it fires on 6% of readouts at 0.11 mean layers; sharing one token — always `.py` — on 83% at 8.17. Name *length* was the first hypothesis and the data rejected it: $r = 0.183$ over 120 observations, and the four-token name outscores the eleven-token one.

Correcting it symmetrically — dropping shared tokens from target and decoy alike — tightened `at_path` and left `at_sentence` unchanged. We had expected it to rescue the result; a one-sided correction appeared to, and was not licensed.

What the decoy caught

60 readouts, each scored against the target and against another case's target



Five of the six near-ties are `cn1_blueprint.py` against `llamafy_baichuan2.py`, in both directions — unrelated files in unrelated repos.
The control is doing its job: it flags the rows where a hit is a shared subword rather than the file. Scoring on distinctive tokens only is the fix, and it needs a re-run, because the probe recorded how many layers hit and never which tokens they hit on.

Figure 5: Why the decoy is not decoration. Most readouts sit against the vertical axis. The six in the tie band are traceable to a two-character token.

7.d. Provenance of the numbers

Two exports and two runs sit behind this section, and they measure different things:

	cases	claims	cut taken from	raw readout
first export	30	17	message	complete — top-20 ids at 39 layers

	cases	claims	cut taken from	raw readout
second export	75	34	reasoning_content	complete — 149 readouts, all 75 cases

The first export only found claims whose path also appeared in the visible message, so its cut sat in the wrong text. Its figures are retained above only where they describe the transport or the scoring defect — neither of which depends on the cut being early.

The second export reads the field the miner actually found the claim in and is $2.5\times$ larger; it is the one the headline table is computed on. Its lens run was pruned by the compute provider at row 145 of 150, but the probe appends each readout as it is produced rather than writing at the end, so the rows survived; the two remaining cases were scored by a short resume run and merged. The resume reproduces the seeded decoy pairing exactly, because the probe iterates the whole case list and skips rather than slicing it — slicing would reseed the shuffle and re-pair every target.

8. WHERE THIS SITS, AND WHAT IT DISAGREES WITH

Perturbation tests measure self-consistency, and the field knows it. The dominant way to test chain-of-thought faithfulness is to corrupt the reasoning and see whether the answer moves (Lanham et al., 2023; Turpin et al., 2023). Parcalabescu & Frank (2024) argue that this measures the model’s tendency to produce a desired output under perturbation rather than faithfulness in Jacovi & Goldberg (2020)’s sense, and the objection is well taken: an unchanged answer is scored unfaithful whether the model ignored the corrupted step or noticed and silently repaired it.

The setting here sidesteps that argument rather than resolving it. A **relocation** — the model abandoning the file it was asked about and naming a different one, mid-reasoning — is a mid-stream revision that the model produces *spontaneously*. Nothing is corrupted, so there is no perturbation whose effect has to be disentangled from the model’s response to being perturbed. And because the corpus is built from real advisories, the fix commit says whether the revision was **right**. That is the property a GSM8K-style perturbation study cannot have: it can ask whether the answer moved, not whether the reasoning went somewhere true.

Two specific disagreements with how this is usually done.

Accuracy conditioned on a behaviour is a selection artifact waiting to happen. A natural analysis is to split traces by whether the behaviour occurred and compare accuracy. The trap is that the behaviour and the truncation budget are correlated: long traces are both likelier to contain a given phrase and likelier to be cut off. A study reporting that a self-correction marker predicts 100% accuracy, against a corpus whose overall accuracy is far lower, has almost certainly conditioned on completion. We report the analogous split (21 correct against 14 wrong, § *The follow-up*) with the labelling rule

stated and both rules' outcomes given, because the rule moved the balance from 21/14 to 27/11 on its own.

A resampling test without a matched base rate is not a test. Regenerating from a context that historically preceded a behaviour, and finding the behaviour recurs, establishes very little unless the same procedure is run from matched contexts that did not. That is the entire content of our control export — same run, no relocation in the step, matched on prefix length and final character — and getting it right took three attempts, each of which produced a confident number first (§*The follow-up*, Q3).

Attention to a delimiter is the null hypothesis, not the finding. Reasoning models attend heavily to `\n` and to structural tokens near the start of the context; this is the well-documented attention-sink pattern and it appears in essentially every decoder. Our own analogue of that mistake was cheaper to catch and more embarrassing: a probe that scored AUC 0.894 **at the embedding layer**, which no amount of mechanism could explain, because no block had run. It was punctuation.

On the instrument. The Jacobian readout tested here is a transport in the family of the logit lens (nostalgebraist, 2020) and the tuned lens (Belrose et al., 2023). Our negative is about *this* readout at *this* sample size and is not evidence that the information is absent — the ask-the-model baseline recovers the file in 18 of 30 claims, so something at that point determines it.

On the task. Vijay et al. (2026) establish two facts this corpus inherits. First, localization difficulty follows structure rather than severity, and performance *collapses* on vulnerabilities spanning five or more files — which is the backdrop for our Q1 failure: relocated filenames are nested inside repositories, only 4 of our 15 repositories host two distinct relocations, and a within-repository decoy therefore has almost nothing to score. Second, their §C.2 shows that adding a three-phase strategy to the system prompt, with no retraining, moves File F1 from 0.223 to 0.2313 and structural exploration from 10.2% to 17.3%. Our Q4 result is the same phenomenon at finer grain and with a causal trace: a consolation clause in the prompt drives stated relocations out of the submitted output at 17.2% against 0.5% (odds ratio 39), and one transcript records the model resolving to cite the right file and then being redirected mid-sentence. Prompt-level instructions are not a nuisance parameter in agentic localization; they are a load-bearing part of the measured behaviour.

Zhang et al. (2026) separate exploration from solving into a dedicated subagent rewarded on patch-derived file and line F1, and two of their methodological choices are ones we arrived at independently and for the same reasons. They report **instance-wise averages rather than micro-averages** “so repositories with large patches do not dominate” — which is precisely our per-claim rather than per-mention unit, and the correction that shrank our first published figure when one claim turned out to be restated seven times. And they note their evaluation “intentionally rewards recovery of edited locations; it may under-credit supporting evidence such as tests, callers, configuration files, or neighboring implementations that are useful to an end-to-end solver but not modified by the reference patch.” Our gold has exactly this

shape, and it is why § *The follow-up* reports the strict path-match labels and the looser basename labels side by side instead of choosing one silently.

9. THE FOLLOW-UP, AND WHAT IT RETURNED

The divergence between stated and submitted location is visible in the transcript, after the fact. The question this dataset is positioned to ask is whether it is visible **in the activations, and before the output is produced**.

All of the questions below have now been attempted, and none came back the way it was proposed. Question 1 turned out to be unanswerable on this corpus, for a reason worth stating precisely. Question 2 is null, against a null distribution that had to be measured rather than assumed. Question 3 was added during the follow-up. Question 4's premise did not survive reading the traces.

Each is reported with the number that was wrong first. That ordering is not modesty — across the follow-up, **five separate artifacts produced a publishable-looking result, and every one of them pointed in the direction the hypothesis wanted**: a repository detector reading as a filename probe, a delimiter detector reading as an event probe, a prompt-length shortcut, a per-mention count inflating a per-claim p-value, and a permutation null built at the wrong unit. None was caught by finding the result implausible. Each was caught by a control built in advance to catch that class of error, which is the only method that works when the artifact and the hypothesis agree.

Concretely, the corpus supplies per trace a triple — stated location, submitted location, true location — with 218 relocations mined and 6 where a stated location never reaches the output.

The model that relocates often enough to study is the 35B, which does not fit on the local card, so both extractions ran on a rented A100 with the weights offloaded to system RAM. That constraint shapes what is affordable: a full residual extraction is ~ 45 minutes of GPU time per condition, and the provider prunes long sessions without warning — it took one run mid-flight during this work. Every extraction therefore writes incrementally and is pulled as it grows, so a prune costs a tail rather than a run.

Four questions, in increasing order of interest.

9.a. Q1. *Is the submitted answer decodable before it is stated?*

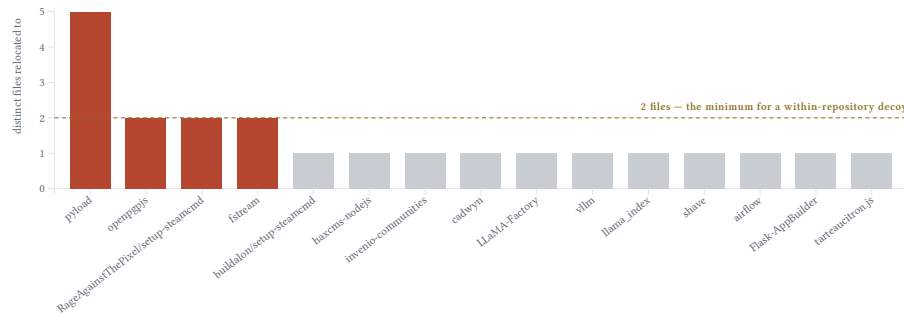
Attempted. The corpus cannot answer it, and finding out why is the more useful result.

A second extraction saved the residual stream itself — 150 readouts, 41 layers, 2,048 dimensions — and a nearest-centroid readout scored 20 of 24 claims over its decoy at the last layer ($p = 0.0015$), rising monotonically with depth.

That number is an artifact. Relocated filenames are nested inside repositories: 20 of 21 distinct basenames occur in exactly one repo, and 71 of 75 decoys name a file from a

different project than the one being read. The residual encodes which project it is reading, because the context *is* that project. Restricting the decoy to a different file from the same repository removes the shortcut and leaves **two scoreable claims**, because only 4 of 15 repositories ever host two distinct relocations.

The question is well-posed and this corpus is the wrong instrument for it. A successor has to be built with within-repository relocation pairs rather than filtered into having them.



Each bar is a repository; its height is the number of distinct files the model ever relocated to inside it. Only the four above the dashed line can supply a same-repository decoy at all. With filenames nested inside projects this way, “prefers the target over the decoy” is satisfiable by recognising the repository — which every method under test can do, because the context is that repository’s source.

9.b. Q2. Do divergent traces look different at the point of divergence?

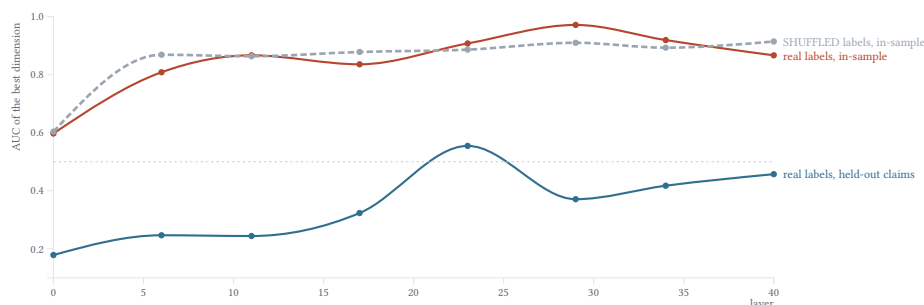
Attempted, and null.

Logistic regression on the residual at the earlier cut, predicting whether the relocated file is in the fix commit, leave-one-claim-out with PCA fitted inside each fold and a claim-level permutation null. 35 claims, 21 correct against 14 wrong. No layer clears its null; the best is AUC 0.517 against a null mean of 0.435 ($p = 0.30$). The null sits near 0.42 rather than 0.5 because of the class imbalance and skipped folds — which is precisely why it has to be measured rather than assumed, since read against 0.5 the same number would have looked like a weak positive.

What the null is worth, stated as power rather than as prose. The null’s 95th percentile at the best layer is **0.676**. That is the detection threshold this design actually had: at 35 claims, no effect below AUC 0.676 could have been called significant however real it was. 0.676 is a large effect, so this is a weak null — it bounds the effect, it does not exclude one. Any successor needs claims in the hundreds, and the constraint is not compute but corpus: 218 relocations exist, and the 75 used here were what one A100 session could extract.

A separate sweep asked what a single dimension is worth here, running the familiar recipe — rank every unit by ROC-AUC, take the best, check Cohen’s d , observe it fails to generalise — and adding the step usually skipped: the same procedure on shuffled labels. Mean in-sample AUC was **0.847** with real labels and **0.852** with random ones,

held-out **0.349**, with Cohen’s d reaching 0.65 on provable noise. At 2,048 dimensions and 35 claims the best-looking unit is the best of 2,048 draws, and “high AUC, large effect, fails to generalise, therefore polysemanticity” is a conclusion this data reaches with no signal present at all.



Three curves over depth: real labels in-sample, the same procedure on shuffled labels in-sample, and real labels scored on held-out claims. The first two are the same number. The recipe that ends in “polysemanticity” reaches that conclusion here with nothing present.

One detail of that sweep is itself a lesson. The first version drew a fresh claim-split for each permutation while scoring the real labels on one fixed split, which compares a single draw against an average over draws. Pinning the split changed mean Cohen’s d from 0.97 to 0.65 and made the real and shuffled curves converge — the corrected version is the stronger result, and the uncorrected one would have been the more impressive-looking figure.

9.c. Q3. Is the relocation event decodable, before the sentence that carries it?

Added during the follow-up, because it is the question the chain-of-thought literature asks of backtracking — and a relocation is backtracking with an answer key. It needs negatives, so a second export emits one matched control per relocation: a sentence boundary in the *same run*, in a step containing no relocation, chosen to match the positive’s prefix length.

The matching is the entire experiment, and it took three attempts to make it a control rather than a tell. Each failure produced a confident-looking number first:

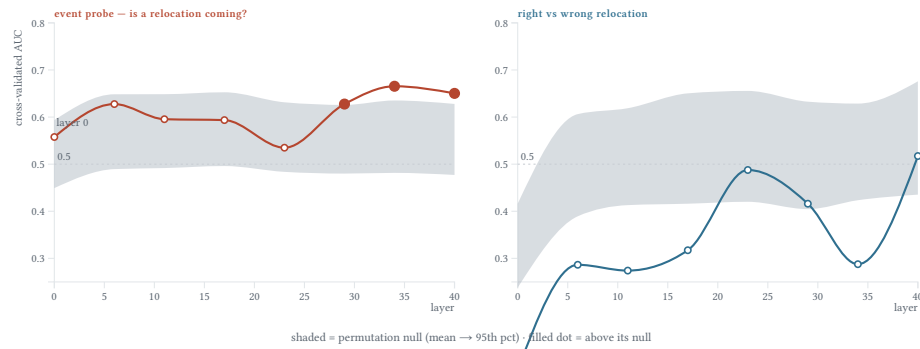
- **Prompt length.** Relocations arrive late in long traces. Matching the reasoning prefix to a median of 14 characters was not enough, because the prompt also carries the conversation and controls sit at earlier steps (20.6 messages against 27.3). Token count alone separated the classes at **AUC 0.62**. Capping the within-pair token gap at 30% drops that to 0.498.
- **Punctuation.** The first probe scored **AUC 0.894 at layer 0** — the embedding output, before a single transformer block has run, where no separation can be a fact about computation. The two exports cut on different conventions: relocation prefixes end on the punctuation mark and control prefixes kept the trailing space, so relocations ended on . 45 times of 75 and controls never did, ending on a space 28 times instead. The probe was a delimiter detector.

- **The residue of the fix.** Cutting both on one convention was necessary and insufficient: usable relocations still ended on . 71% of the time against 35% for controls, worth about 0.68 to a classifier reading the last token alone. The export now stratifies on final character before matching on length.

Layer 0 is the cheapest diagnostic in this report and the one that caught the worst error. A probe that separates classes before any block has run is reading its input, not the model.

The corrected result: suggestive, and not a finding. On the 23 pairs that are both length-balanced and final-character matched — where the length shortcut is worth 0.499 — the probe behaves the way a real signal should and still fails to clear the bar:

layer	AUC	null 95th pct	p
0 (embeddings)	0.558	0.594	0.103
29	0.629	0.628	0.053
34	0.665	0.635	0.033
40	0.650	0.630	0.040



Each panel draws its own permutation null as a band (mean to 95th percentile) for the identical procedure; a filled dot is a layer above its null. The event probe rises out of its band late and starts inside it at layer 0 — the right shape. Right-vs-wrong never leaves its band. Reading either curve against 0.5 instead of against its band would give the wrong answer in both panels.

Layer 0 is now non-significant, which is the diagnostic passing: nothing is separable before the model computes anything, and what separation exists appears late. That is the right shape. But eight layers are eight tests, and Benjamini–Hochberg (Benjamini & Hochberg, 1995) at $q = 0.05$ requires the smallest p below 0.00625. The smallest is 0.033. **Nothing survives correction**, and at 23 pairs the detection threshold was AUC 0.635 — so a real late-layer effect of ordinary size would have been invisible regardless.

The correct reading is that the experiment is now *well-formed* and *underpowered*, which is a better place to be than the version that returned AUC 0.894 at layer 0. The

design is reusable: 218 relocations exist against the 75 extracted here, and the matched-control export is the part that took three attempts to get right.

9.d. Q4. *Is the never-submitted set mechanistically distinct?*

Answered, and the premise was wrong. It is neither a suppression story nor an error story: the rate is 17.2% under a prompt carrying a consolation clause and 0.5% without it ($p = 0.00016$, odds ratio 39), and one trace records the model deciding to cite the right file and then being redirected by the instruction. A mechanistic account was not needed; reading the six traces was.

The value of the setting is that Q1 can be scored against an answer key that neither the transcript nor the output produced.

10. LIMITATIONS

- 37 distinct claims is a small denominator. The per-claim precision of 40.5% has a wide interval, and the honest reading is “below baseline, not usefully estimated”.
- Three tasks dominating the mention population means the corpus is not diverse at the trace level, whatever the task count suggests.
- Relocation is one operationalisation of stated/acted divergence and the narrowest one. It does not capture a run that reasons correctly and submits correctly for a stated reason that is false.
- The agreement calibration (49.8% at 3 of 3) is measured on this corpus and this specialist. It is a local baseline, not a general one.
- Gold ranges come from fix-commit hunks with context, so line-level scoring inherits whatever the maintainer included in the diff.
- The pilot has ~30 distinct claims from 15 repositories. That is enough to establish the baseline result ($p = 0.0003$) and not enough to distinguish a weak lens effect from none.
- The lens arm covers all 75 cases. The main run was pruned by the provider at row 145 of 150; the rows survived because the probe appends each readout as it is produced rather than writing at the end, and the two remaining cases were scored by a short resume run and merged. The decoy pairing is identical across both because the probe iterates the full case list and skips rather than slicing it.
- The baseline lets the model generate up to 2,400 tokens before answering, while the lens reads one activation. The comparison is deliberately unfair in the direction that makes the lens negative safe, and correspondingly weak as evidence that nothing is in the residual stream.
- Lens scoring counts a hit when any token of a filename enters the top 20, which is permissive. Shared tokens are now removed from target and decoy alike, which fixes the collision the control exposed, but the rule is still “any token at any layer” and a hit is not a rank.
- The emergence curve is a per-layer *hit rate over claims*, not a confidence trajectory. It says where the token enters a top-20, not how strongly. Layer 38 is the model’s own output distribution, so the logit lens arriving there is expected and is not evidence about the transport.

- The seeded decoy controls for filename *plausibility* but not for repository identity: 71 of 75 decoys name a file from a different project than the one being read. Every target-vs-decoy figure in this report therefore reports a floor that is too low. This does not change the lens result, which was null against an over-generous floor, and it moves the baseline’s load-bearing number from the 18-versus-3 gap to the 18-of-30 exact-path hit rate.
- Because relocated filenames are nested inside repositories — 20 of 21 occur in exactly one repo, and only 4 of 15 repos host two — a within-repository version of the decoy leaves too few scoreable claims to test. The question “does the residual identify the file rather than the project?” is **not answerable on this corpus**, and a successor corpus has to be built for it rather than filtered into it.

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